

Project Overview.

Wajha is an AI-powered platform built specifically for Palestinian students and graduates searching for scholarships and funding opportunities. It brings together verified grant listings, deadlines, and eligibility details in one place, making it easier to discover relevant opportunities, prepare strong applications, and track progress from submission to decision. Our platform includes Smart Grant Matching, AI-Assisted Application Support, Deadline Tracking, and Application Status Management. With a community-first approach and modern AI technology, we help students cut through scattered, overwhelming information and focus on the opportunities that truly fit them. **Wajha** bridges the gap between talented Palestinian students and global funding opportunities, one application at a time.

Problem Solved

Palestinian students and graduates face a uniquely difficult path to higher education funding: scholarship information is scattered across dozens of websites and social media pages, eligibility criteria are often buried in long, complex documents, and application deadlines are easy to miss amid daily disruptions and limited access to stable internet. Many qualified students miss opportunities not because they're unqualified, but because they never found the right grant in time, didn't know how to present their profile effectively, or lost track of where they stood in the process. **Wajha** solves this by centralizing grant information, using AI to match students with opportunities suited to their background, simplifying complex eligibility requirements into clear guidance, and giving students a structured way to track every application from "saved" to "accepted" or "rejected" so no opportunity slips through the cracks.

Key Features of the Platform

- **Grant Database**
A centralized, regularly updated collection of scholarships and funding opportunities with deadlines, eligibility criteria, and sources.
- **AI-Powered Matching**
Each student's profile is analyzed against available grants to surface the most relevant opportunities, with AI-generated explanations of why each one fits.
- **Application Assistance**
AI helps students understand complex eligibility requirements and draft tailored application materials, such as motivation letters.
- **Application Tracking**

Every saved grant moves through clear status stages (Saved, In Progress, Submitted, Under Review, Accepted, Rejected), so students always know where they stand.

- **Deadline Alerts**

Automatic reminders as deadlines approach for grants that haven't been submitted yet.

- **Role-Based Access**

Secure login with session-based authorization.

How **Wajha** Works

1. User Roles & Access

Students register and build a profile with their academic and personal background.

Visitors can browse general grant listings without an account.

Admin manages grant listings, users, and content.

Secure session-based authentication protects all accounts.

2. Student Profiles

Each user has a profile with field of study, degree level, GPA, languages, country, and an uploaded CV.

Profile page shows saved grants, application history, and current status for each.

3. Grant Management

Admin adds grants with title, organization, description, eligibility criteria, deadline, funding type, and source link.

Grants are tagged by field of study, degree level, and eligible countries.

Full CRUD create, read, update, delete grants (admin-side).

4. Application Tracking

Students save grants they're interested in, moving them through clear stages:

Saved → In Progress → Submitted → Under Review → Accepted/Rejected.

Deadline countdowns shown on every saved grant.

Status changes update instantly via AJAX.

All tracked applications visible on a personal dashboard.

5. Discovery & Search

Search grants by title, field, or organization.

Filter by degree level, country, or deadline range.

AJAX-powered live search.

6. API & Documentation

REST API endpoint returns grant and application data as JSON.

Full GitHub documentation with setup instructions.

Web Scraping Module (Grant Aggregation Engine)

This module keeps GrantPal's grant database fresh without relying solely on manual admin entry. It periodically visits a configured list of trusted scholarship and funding websites, extracts structured grant data, and brings it into **Wajha** for review and publishing.

Purpose

- Reduce manual data-entry workload for admins.
- Keep deadlines, eligibility text, and funding details current.
- Surface new opportunities from official sources as soon as they're published.

Pipeline Stages

1. **Source Configuration** — Admin registers a target site URL, scrape frequency (daily/weekly), and a CSS/XPath selector map for fields like title, organization, deadline, eligibility, and link.
2. **Scheduled Fetch** — A background job (Celery + Celery Beat, or cron) visits each source on schedule and downloads the page HTML.
3. **Parsing & Extraction** — BeautifulSoup / lxml (or Scrapy for larger sites) extracts the mapped fields into a structured record.
4. **Normalization** — Dates parsed into a consistent format, funding amounts standardized, text cleaned (whitespace, HTML artifacts removed).
5. **De-duplication** — New records compared against existing grants by source URL + fuzzy title match to avoid duplicates.
6. **Review Queue** — Unverified scraped grants land in an admin-only "Pending Review" queue rather than going live automatically.
7. **Admin Approval** — Admin edits/approves entries, tags them, and publishes to the live Grant Database.
8. **AI Eligibility Summarization** — Once approved, AI generates the plain-language eligibility summary automatically.

Component	Technology / Notes
Scraping engine	Python: requests + BeautifulSoup for simple static pages; Scrapy for larger multi-page crawls; Playwright/Selenium only for JS-rendered sites
Scheduling	Celery + Celery Beat (or django-cron) for recurring scrape jobs
Storage	Raw scraped payloads stored temporarily; structured data written to a ScrapedGrant staging table (separate from the live Grant table)
Error handling	Per-source try/except with logging; failed sources flagged in the admin dashboard, not silently skipped

Rate limiting & ethics	Respect robots.txt , throttle request frequency per domain, use a descriptive User-Agent, only scrape publicly accessible pages
De-dup logic	Hash of (source_url + title) plus fuzzy title matching (e.g., rapidfuzz) against existing grants

Admin Review Dashboard

- Pending Review tab listing all newly scraped grants with source attribution.
- Side-by-side view of raw scraped text vs. parsed fields for quick correction.
- One-click Approve / Reject / Edit-then-approve actions.
- Source health panel showing last successful scrape time and failure count per source.

Data Model Addition: **ScrapedGrant** (staging)

Field	Type	Description
source	FK → GrantSource	Which configured site this record came from
raw_title	TEXT	Title as extracted, pre-cleaning
raw_html_snippet	TEXT	Original HTML fragment, kept for admin verification
parsed_data	JSON	Normalized fields: title, org, deadline, eligibility, funding_type, link
status	ENUM	pending / approved / rejected / duplicate
scraped_at	TIMESTAMP	When the record was captured
reviewed_by	FK → User (admin)	Admin who approved/rejected the entry

Role of AI

AI is not a side feature in **Wajha** it's the layer that turns a static, scraped grant list into a personalized guidance system:

- **Smart Matching** — After basic filtering by field, degree level, and country, AI reviews the student's full profile against remaining candidates and ranks them by genuine fit, not just keyword overlap.
- **Plain-Language Explanations** — AI generates a short, clear explanation of why each matched grant suits the student.
- **Eligibility Simplification** — AI summarizes long, complex grant requirement pages (including freshly scraped listings) into simple, scannable points in Arabic or English.

- **Application Support** — AI helps draft a first version of a motivation letter tailored to the grant and the student's background, and reviews CVs for relevant improvements.
 - **Gap Analysis** — AI identifies what's missing from a student's profile so they can prepare ahead of future opportunities.
 - **Scraped-Data Tagging** — AI assists admins by auto-suggesting field-of-study, degree-level, and country tags for newly scraped grants before publishing.
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Project Requirements

1. Functional Requirements

- User Registration & Login — Secure sign-up and authentication.
- Profile Management — Create/edit student profile with academic background, GPA, languages, CV.
- Grant Management — Add, edit, delete, view grants (admin).
- Automated Grant Scraping — Configure sources, run scheduled scrapes, review scraped entries before publishing.
- AI Matching — Rank grants by fit against student profile with explanations.
- Application Assistance — AI-drafted motivation letters and CV review.
- Application Tracking — Move saved grants through status stages.
- Search & Filter — By field, degree level, country, or deadline.
- API Endpoint — GET /api/grants/ returns all grants as JSON.
- About Us Page — Platform mission, team, and story.
- Responsive Design — Mobile-friendly on at least 3 pages.

2. Technical Requirements

- Backend Framework — Django (Python)
- Database — MySQL
- Frontend — HTML, CSS (Bootstrap 5), JavaScript
- Web Scraping — BeautifulSoup / Scrapy + Celery + Celery Beat for scheduling
- AJAX Integration — Status updates, search, save actions without page reload
- RESTful API — JSON endpoints for grant and application data
- Authentication & Security — Hashed passwords, CSRF protection, session-based authorization
- Deployment — AWS EC2
- Version Control — GitHub with README and setup docs

3. Non-Functional Requirements

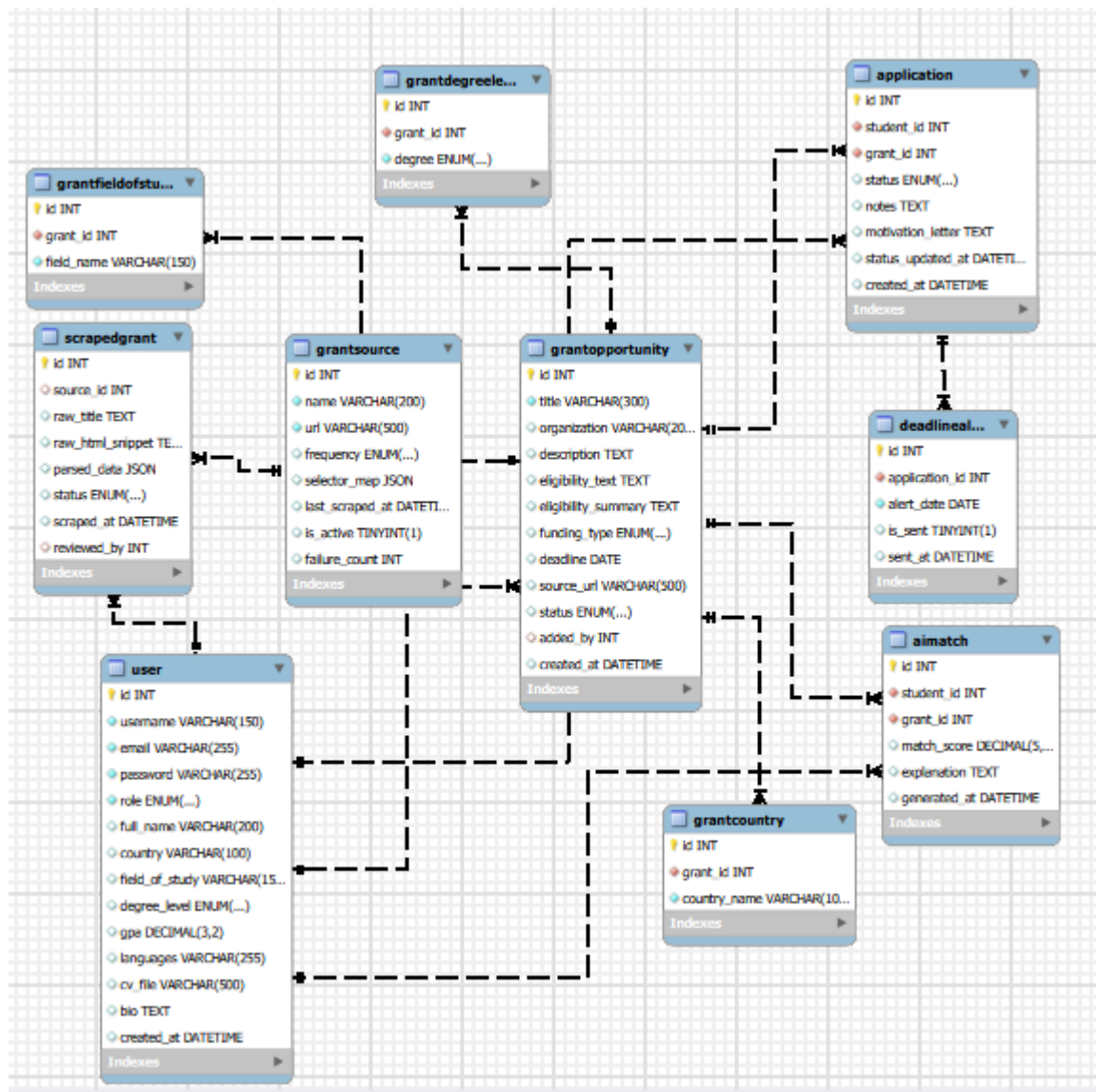
- Usability — Clean, modern UI intuitive for all users.
- Performance — Optimized ORM queries; scrapers run asynchronously without blocking the app.
- Reliability — Stable with minimal downtime; scraper failures isolated per-source.

- Maintainability — Clean MTV architecture, documented code, admin-configurable selector maps (not hardcoded).
- Compatibility — Works across Chrome, Firefox, Safari, mobile browsers.
- Ethical Scraping — Respect robots.txt and site terms of use; throttle requests; only collect public, non-personal data.

Tech Stack

Layer	Technology
Backend	Django (Python)
Database	MySQL
Frontend	HTML5, CSS3, Bootstrap 5
Scraping	BeautifulSoup4 / Scrapy, requests, lxml
Task Scheduling	Celery + Celery Beat (or Redis-backed queue)
AI Layer	Anthropic API (Claude) for matching, summarization, and drafting
Auth	Django Sessions + hashed passwords, CSRF protection
AJAX	Vanilla JavaScript / Fetch API
Deployment	AWS EC2 (Nginx + Gunicorn)
Version Control	GitHub

ERD Diagram



Core Entities (per your ERD)

Entity	Key Fields
User	username, email, password, role, full_name, country, field_of_study, degree_level, gpa, languages, cv_file, bio
GrantOpportunity	title, organization, description, eligibility_text, eligibility_summary, funding_type, deadline, source_url, status, added_by

GrantSource	name, url, frequency, selector_map, last_scraped_at, is_active, failure_count
ScrapedGrant	source_id, raw_title, raw_html_snippet, parsed_data, status, scraped_at, reviewed_by
Application	student_id, grant_id, status, notes, motivation_letter_text, status_updated_at
DeadlineAlert	application_id, alert_date, is_sent, sent_at
AIMatch	student_id, grant_id, match_score, explanation, generated_at
GrantFieldOfStudy	grant_id, field_name
GrantDegreeLevel	grant_id, degree (ENUM)
GrantCountry	grant_id, country_name

Relationships

Application → User (Many-to-One),
 Application → GrantOpportunity (Many-to-One)
 ScrapedGrant → GrantSource (Many-to-One)
 GrantOpportunity ← approved ScrapedGrant (One-to-One, optional)
 AIMatch → User + GrantOpportunity (Many-to-One each)
 DeadlineAlert → Application (Many-to-One)
 GrantOpportunity ↔ GrantFieldOfStudy / GrantDegreeLevel / GrantCountry
 (One-to-Many tag tables)

GrantPal
Browse Matches Dashboard 27

Find scholarships built for you — one application at a time.

Verified grants, deadlines & eligibility in one place. All matches you to the ones that truly fit.

Got matched → Browse all grants

1,240+ verified grants 38 countries All degree levels

Quick filter:
Field of study Degree level Country Deadline
Sort: Deadline

Latest opportunities

Masters
Fulbright Foreign Student Program
U.S. Dept. of State
18 days left
Save +

PhD
DAAD Study Scholarship
DAAD - Germany
42 days left
Save +

Masters
Chevening Scholarship
UK Government
90 days left
Save +

Bachelor
Erasmus+ Mobility Grant
European Commission
30 days left
Save +

PhD
Australia Awards
Govt. of Australia
120 days left
Save +

Masters
Türkiye Bursları
Govt. of Türkiye
9 days left
Save +

GrantPal
live
Browse Matches Dashboard 27

Filters
Degree level
☒ Masters
☐ PhD
☐ Bachelor
Country
Germany X
Field of study
Any
Deadline within
≤ 60 days
Clear all

27 grants match — sorted by deadline
Sort · List

DAAD Study Scholarship
DAAD - Germany · fully funded
Masters CS 42 days left
Save +

Heinrich Böll Foundation
HBS - Germany · monthly stipend
Masters PhD 12 days left
Save +

TU Munich Merit Award
TUM - Germany · tuition waiver
Masters CS 70 days left
Save +

KAAD Scholarship
KAAD - Germany
88 days left
Save +

Load more results ↓

03 · Grant Detail

6 GrantPal

← Back to results

Masters · Fully funded

DAAD Study Scholarship 2026

DAAD — German Academic Exchange Service · Germany

AI Eligibility, simplified

- You must hold a Bachelor's (you do ✓)
- Min. GPA 3.0 — yours is 3.6 ✓
- German not required; English-taught program ✓
- Need: 2 recommendation letters, motivation letter

Generated from the official eligibility page · also available in العربية

About this grant

Eligibility criteria

Computer Science Engineering Open to Palestine

Deadline

15 Aug 2026

⌚ 42 days left

Save & track →

+ Draft my application

[Visit official source ↗](#)

At a glance

Funding · Full tuition + €934/mo

Duration · 12–24 months

Level · Masters

Languages · English

04 · AI Matching Results + Explanations

6 GrantPal

Browse Matches Dashboard ST

12 grants matched to your profile

Ranked by genuine fit — CS · Masters · Palestine · entrepreneurship experience

Update profile ▶

96%

strong fit

DAAD Study Scholarship

Masters

fully funded · ⌚ 42 days

Why it fits →

Your CS background + 3.6 GPA clear the bar, and DAAD funds Palestinian Masters students in English-taught programs. Your startup experience strengthens the motivation letter angle.

Save +

View

89%

strong fit

Chevening Scholarship

Masters

⌚ 90 days

Why it fits →

Chevening values leadership — your entrepreneurship experience maps directly to their selection criteria. Also opens a business-track angle beyond pure CS.

Save +

View

74%

good fit

Erasmus Mundus Joint Masters

Masters

commission · ⌚ 30 days

Why it fits →

Field matches, but a few programs ask for a B2 language certificate you don't have yet — see Gap analysis on your profile.

Save +

View

05 · AI Application Assistant (motivation letter)

6

GrantPal

AI Assistant

Pulling from your profile

CV · uploaded ✓

Field · Computer Science

GPA · 3.6

Experience · startup founder

Grant · DAAD (eligibility loaded)

Tone

Formal

Personal

Concise

Add a focus

e.g. emphasise my open-source work...

+ Generate draft

Review my CV instead

Draft for: DAAD Study Scholarship

Regenerate

Copy

Save to grant

Motivation letter — draft

Dear DAAD Selection Committee,

+ highlighted lines pulled from your profile — edit freely

AI draft — review & personalise before submitting. You're always the author.

06 · Student Dashboard (tracked applications)

6 GrantPal

[Browse](#)
[Matches](#)
[Dashboard](#)

ST

My applications

7 tracked

2 deadlines this week

SAVED · 2

Türkiye Bursları

9 days

Australia Awards

120 days

IN PROGRESS · 1

DAAD Study Scholarship

42 days - letter drafted

SUBMITTED · 1

Chevening

sent 3 days ago

UNDER REVIEW · 1

Erasmus Mundus

interview pending

ACCEPTED · 1

DAAD EPOS

offer received

REJECTED · 1

Gates Cambridge

not selected

Drag cards between stages - status saves instantly

07 · Student Profile / Onboarding

6 GrantPal

Step 2 of 3 · Build your profile

1

2

3

Tell us about your studies

The more we know, the better AI can match you.

Field of study

Computer Science

Degree level

Masters ▾

GPA

3.6 / 4.0

Country

Palestine ▾

Languages

Arabic × English × + add

Upload your CV

1 Drop CV here or browse — PDF, max 5MB

← Back

Continue →

+ AI Profile gap analysis

To unlock more matches, consider adding:

- A recommendation letter
- An English certificate (IELTS/TOEFL)
- 1–2 portfolio links

Prepares you for future grants — not just today's.

Profile strength

72% — looking good!

08 · Admin — Grant Management (CRUD)

6 GrantPal

ADMIN

Grants

Users

Applications

API & docs

Grants

1,240 total

Search grants...

+ Add grant

TITLE	ORGANISATION	LEVEL	COUNTRY	DEADLINE	ACTIONS
DAAD Study Scholarship	DAAD	Masters	Germany	15 Aug	Edit · Delete
Chevening Scholarship	UK Govt.	Masters	UK	2 Nov	Edit · Delete
Fulbright Foreign Student	US State Dept.	Masters	USA	10 Jul	Edit · Delete
Erasmus Mundus Joint	EU Commission	Masters	EU	30 Jun	Edit · Delete
Türkiye Bursları	Govt. of Türkiye	Masters	Türkiye	20 Feb	Edit · Delete

Showing 1–5 of 1,240

< Prev

Next >

" + Add grant" / "Edit" opens a form: title · organisation · description · eligibility · deadline · funding type · source link · tags (field, level, countries).